

Jitsi360: Using 360 Images for Live Tours

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ABSTRACT

In this poster, we present a system for sharing immersive 360 images for live tours. While the sharing of prerecorded 360 video and images is becoming commonplace, there have been fewer systems presented that support live-sharing of 360 images. We customised a video conferencing platform to enable dozens of people to see the same 360 content together while having a live call. We describe our system and pilot user study results from using it for a virtual guided tour. Compared to sharing non 360 images on Zoom, our system was felt to be more immersive, enjoyable, and easy to use.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Over the last two years, virtual tourism has become increasingly popular, especially due to the limitations placed on physical travel to different locations due to the COVID-19 pandemic. Many tour businesses have switched to offering virtual online tours for their customers. For example, AirBnB Online Experiences¹ offers a platform for tour guides to provide virtual online experiences where typically a handheld mobile phone is used to capture and stream the tour through a video conferencing system such as Zoom. The limitation of this approach is that viewers are becoming passive participants in the tour and are limited to what they see through the presenter camera.

In our research, we have created a platform, Jitsi360, that enables people to give live tours using 360 images or videos of locations, with live audio and video connection. The main novelty of our system is that it is based on an existing video conferencing platform, and so it is secure and scalable to dozens of simultaneous users. We also support features designed for providing tours, such as a remote pointing, view sharing, and content loading. A second novel contribution of this poster is that we report on a pilot trial with the platform and in particular comparison to non-360 image sharing to explore how immersive users found the experience.

2 RELATED WORK

Our work is inspired by previous work that has explored the use of 360 video cameras for enabling independent viewpoint sharing between the presenter and the viewer. [1] presented JackIn where a head-worn camera is used to stream a 360 video from a first-person view, however, this supported only a single remote viewer. Other examples of research using 360 video to enable remote users to look around a guides location include SharedSphere [2], MM360 [5], and Bewithme [6]. However, these systems can't support many simultaneous users.

There are several commercial systems for live-streamed 360 video tours. For example, Avatour² developed a system that uses a hand-

held 360 camera with a mobile phone to live-stream to a small group of viewers. However Avatour does not support remote pointing which makes it difficult for people on a virtual tour to talk about their surroundings.

Static 360 content has been used in replaying sport [4], preserving cultural heritage [3] and promoting mountain walking [7]. However due to being pre-recorded content, these attempts remained different to a real-life tour experience and viewers may feel isolated. To provide a similar experience to a live tour where tourists can look around independently from where the presenter is pointing the camera at or showing, we introduce the Jitsi360 platform. This enables sharing of 360 static content as well as live streaming that enables tour guides to take their customers on a more immersive journey and let them explore the scene independently.

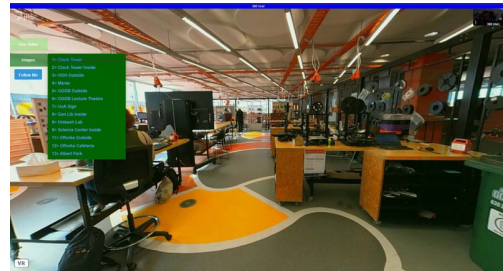


Figure 1: A screenshot of the host view selecting static 360 images from the left menu.

3 THE JITSI360 PLATFORM

The system is built on top of Jitsi which is an open-source video conferencing system for web and mobile devices. We customised the rendering of shared video and images to accommodate 360 content. We used A-Frame to render a sphere with a texture of the shared content and placed a dot on the sphere to represent the direction of where the user is pointing at.

To support virtual tours, This system includes a custom component that enables the host to share pre-recorded static content of 360 images and videos. The host can select this content from a pull-down menu and when this is done, the same content is loaded on all of the viewer screens. This enables the host to give a guided tour and load 360 content at specific points in the tour. Additionally, the host can control the orientation of the viewers to temporarily follow the host orientation.

For the viewer, the user interface is similar to a standard web-based video conferencing interface. Except with they click on a 360-video, they see a full-screen 360 image. They can then click and drag to rotate around the image and see it from different viewpoints. This provides an immersive desktop experience where they may feel they are at the location where the image was taken from. While doing this they can live video feeds from the other connected users and continue to talk them over audio.

The system is running on a Linux-based cloud virtual machine (VM). The camera used by the host in this system is Ricoh Theta V which enables 4K video streaming where the 360 images are streamed in Equirectangular projection format.

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¹<https://www.airbnb.com/s/experiences/online>

²<https://avatour.co/>

4 USER EVALUATION - PILOT STUDY

To explore if the features of the Jitsi360 platform created a more immersive experience than traditional image sharing we conducted a pilot trial in collaboration with a major university. We created a 20 minute scripted virtual open house tour with 360 photos taken from 11 locations around campus (Figure 1). Several students were trained as tour guides, and we ran the tour as a between-groups study with remote visitors being randomly assigned to one of two conditions: Jitsi360 - the 360 image tour, Zoom - a zoom call where the same content was shown using non-360 images.

For the Zoom condition, we created a Powerpoint presentation of the non-360 degree images of the same eleven locations as used in the Jitsi360 tour. We used the same 360 images but cropped them into a rectangular format suitable for Powerpoint. When the Zoom tour was given the guide screen shared the PowerPoint presentation and clicked through the images while presenting the same script as used in the Jitsi360 tour. The intend was to present the same content, with the main difference being that in the Jitsi360 condition, viewers can independently pan around the 360 image.

The virtual tour was given at the same time, with all of the invited remote students being assigned to small groups with a dedicated guide in either the Jitsi360 or Zoom conditions. After introducing themselves the guides would briefly explain how to use the platform (either Jitsi360 or Zoom) and then give the scripted tour. During the tour, participants were free to ask questions about the content being shown. Following the tour, the guide asked the participants to complete a short online survey to capture their feedback. The survey contained Likert scale questions about enjoyment, immersion, ease of use, and interest in using the system to visit another location. Participants were also asked a standard Net Promoter Score (NPS) question³, which is the likelihood that a person would recommend the experience to someone else, and open-ended questions about what they enjoyed most and what could be improved.

The participants were international students invited to a virtual open house from the university. These students were from a variety of countries including China, Japan, India, and others. A total of 13 people completed the final user evaluation, 5 from the Zoom condition, and 8 from the Jitsi360 condition. The participants were asked to answer the following Likert questions on a scale of 1 to 7 (1 = Not very, 7 = Very): Q1: How enjoyable was the experience? Q2: Did you feel like you were really visiting the University? Q3: How easy was it to use the interface? Q4: How much would you like to use the virtual tour system to visit another location?

Table 1 shows the average scores for each of these questions as well as the p-value results of a t-test comparing the means of each question between the two conditions. As can be seen, despite the low number of participants, there was a significant difference in results for Q1, Q2 and Q3 with Jitsi360 scoring significantly higher than Zoom. It is interesting that even though we only used static 360 pictures users still scored the Jitsi360 system at 6.13/7.0 in terms of how immersive they found it. In the Zoom interface users just needed to watch the guide share images, while in the Jitsi360 interface they could click and rotate around the 360 images. Despite this, users rated the Jitsi360 interface as significantly easier to use.

Table 1: User Feedback average results (SD = Standard Deviation)

	Jitsi360		Zoom		T-Test
	Mean	SD	Mean	SD	P-Value
Q1	6.50	0.76	5.80	0.45	0.04
Q2	6.13	0.84	5.20	0.84	0.04
Q3	6.38	0.92	5.00	1.41	0.03
Q4	6.38	1.24	5.80	0.84	0.14

The NPS question was "Q5: How likely is it that you would

³<https://www.netpromoter.com/know/>

recommend this experience to a friend or colleague?", answered on a Likert scale of 1 to 10, with 1 = Not at all likely, and 10 = Extremely likely. For the Jitsi360 the average was a high 9.1/10.0 with an NPS score of 62.5% which is considered great for an NPS result. For the Zoom condition, the average was 8.0/10.0 with an NPS score of 20%, or good. Using a t-test we found a near significant difference between these results ($p = 0.069$). This shows how much more positive users were feeling towards the Jitsi360 experience.

In terms of open-ended questions, people were asked what they liked most about the experience (Q5), and what could be improved (Q6). For both conditions, people enjoyed the content and the friendliness of the guides, but for the Jitsi360 three out of eight people specifically mentioned being able to look around the 360 images. In terms of improvement, six out of eight Jitsi360 users said that nothing needed to be improved, although one user from both conditions asked for more interaction.

5 CONCLUSION

This poster provides one of the first comparisons of using a live 360-image sharing conferencing application compared to non-360 image sharing. Despite the small number of responses, we were able to find a significant difference in how enjoyable and immersive people found using 360 images in a guided experience compared to the non-360 condition. Most interestingly, we were able to show that even just using 360 images was sufficient to create a immersive tour experience (6.13/7.0 rating).

In the future, we will conduct a more formal study with a larger number of users, and more detailed data capture such as the number of times people rotate each image or analysis of the conversation with the guide. In this study, we also didn't test some of the more advanced features such as remote-pointing and shared camera view. Using these features in a formal study would also enable us to capture some interesting behaviour data such as the number of times people pointed at different parts of the image, or how they could maintain awareness of where each person was looking.

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